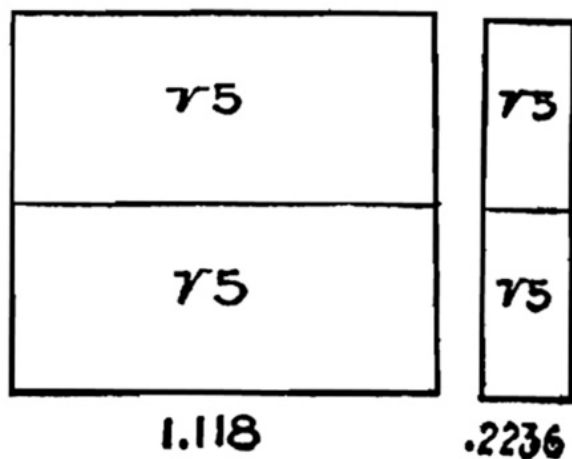
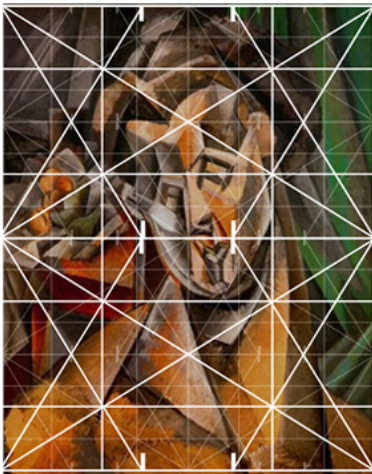




## Composition with Compound Root Rectangles

155/365

How's everyone doing today? Welcome back to more exciting design fun!

If you've read "The Elements of Dynamic Symmetry" by Jay Hambidge ([see Day 14](#)), then this may be familiar to you. Although, if you are not the greatest at math like myself, then you may have overlooked the lessons he demonstrates near the end of his book. It's all about **compound root rectangles**, which will help explain why some **analyzed** canvases have been different than the basic root rectangles.

My mentor, Myron Barnstone, would say that most masters learned geometry and design for 10 years before they even touched the brush to a canvas. I am starting to realize this is true....especially if they fully understood the math and ratios found in Hambidge's book. As artists, we don't have to get to the exact decimal, but seeing how some of the other artists have constructed compound root rectangles will help us in seeing the value and it's easy application. It's just as easy to use as the other grids. Remember, these grids are used for organizing your image and promoting unity and repetition.

Just a side note to anyone who's not too savvy with the ratios. To find the reciprocal of Phi just take 1 and divide by 1.618 = .618 reciprocal. This was difficult for me to understand at first, so I thought I would share. All of the root rectangles have ratios, and all of those ratios can be used to find the reciprocal...which is seen in the illustrations below from Hambidge's book.

## Root Rectangle Ratios

Root Phi = 1.2720

Root 2 = 1.414

1.5 = 1.5

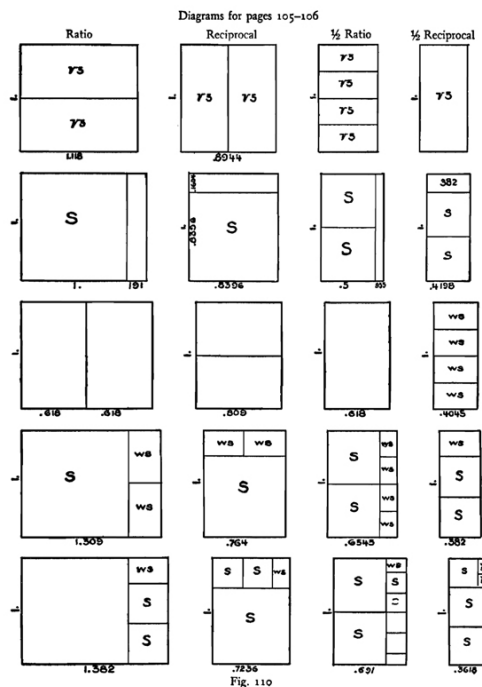
Phi = 1.618

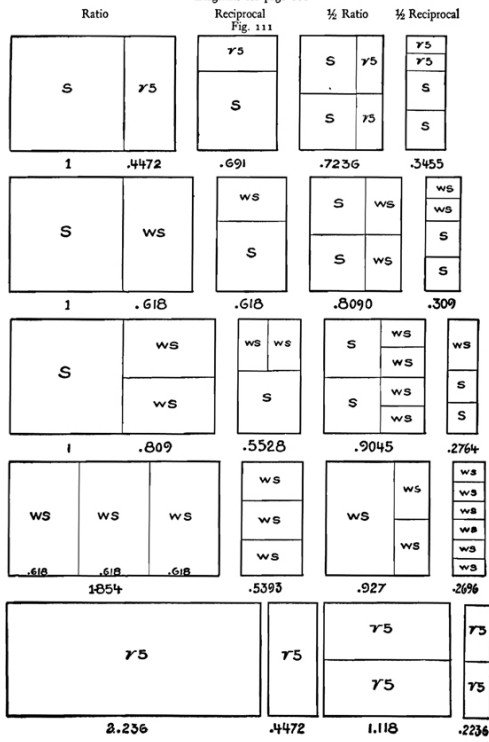
Root 3 = 1.732

Root 4 = 2

Root 5 = 2.236

These illustrations may give you an idea of the variety of compound root rectangles that you can use if you need to. I personally like to keep it simple with the basic root rectangles, but say you have an odd shaped canvas, photo, or wall you want to design within. You can use these compound root rectangles to continue to take advantage of the dynamic symmetry as well as the organization, unity, and repetition it promotes.





### Lesson IX

113

Diagrams for pages 110-114

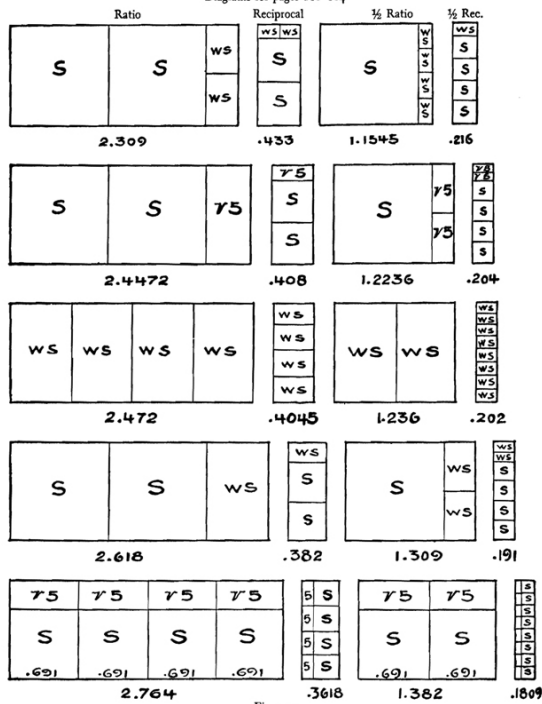


Fig. 115

Diagrams for page 110

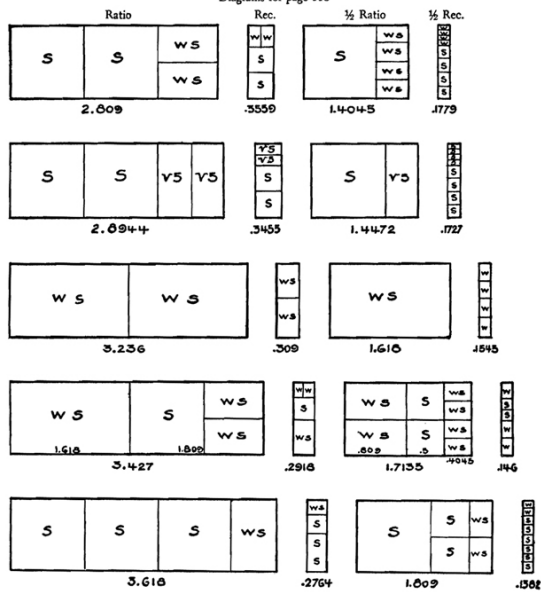
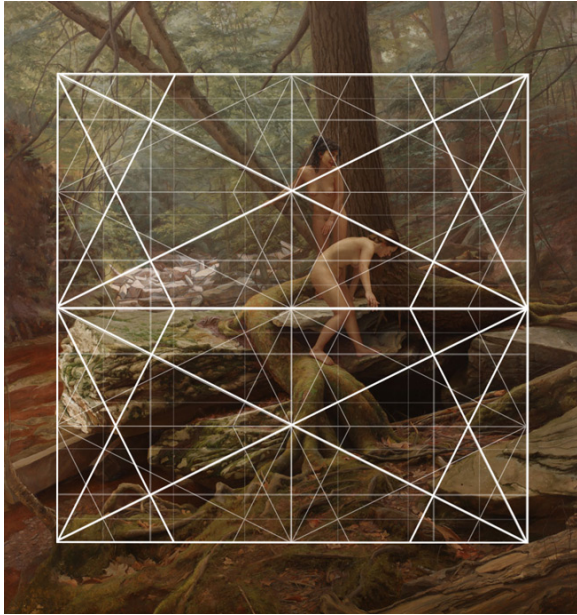


Fig. 116

Here's Giorgio Morandi using two root 5 rectangles stacked and two Phi rectangles at the top.



Here is Arantzazu Martinez using two root 4 rectangles which we all know to equal a square.



Fantin Latour uses two Root 3 rectangles stacked.



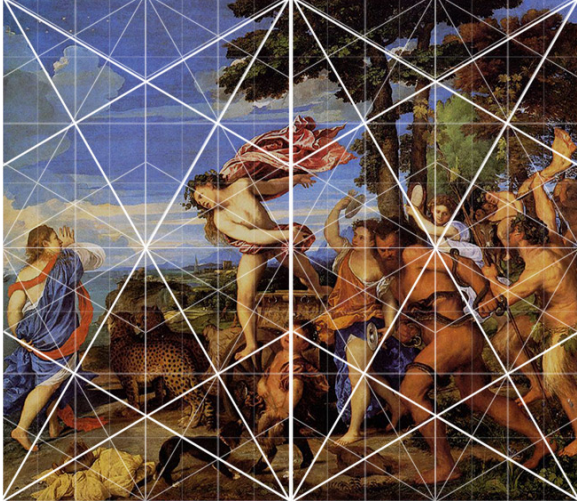
Here's a Johannes Vermeer with stacked Root 3's as well.



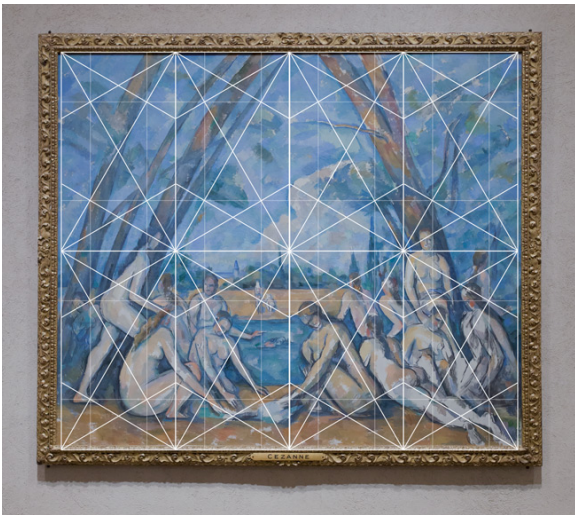
This is a Rubens painting with Root 5's stacked.



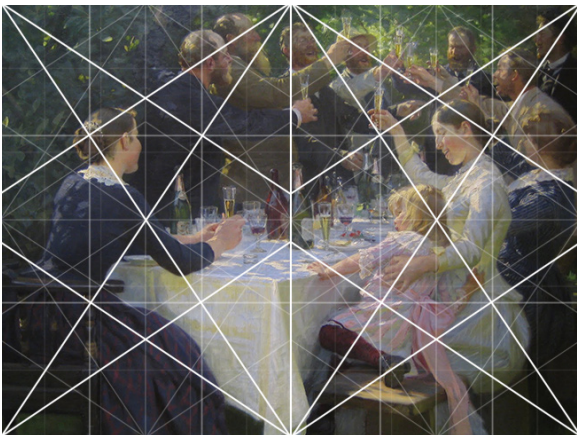
Titian using side by side Root 3's.



Cezanne using two root 3 rectangles.



This is Viggo Johansen using two 1.5 rectangles side by side.



Here's another example of 1.5 rectangles side by side from Caspar David Friedrich.

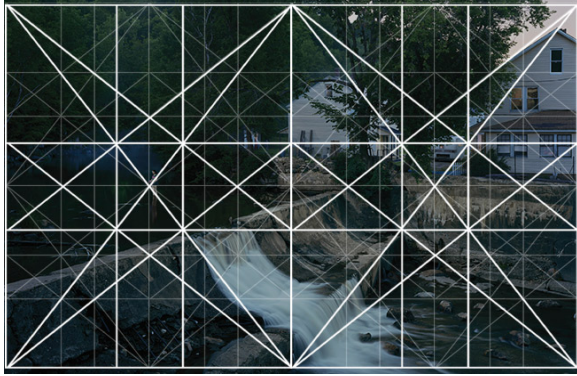
---



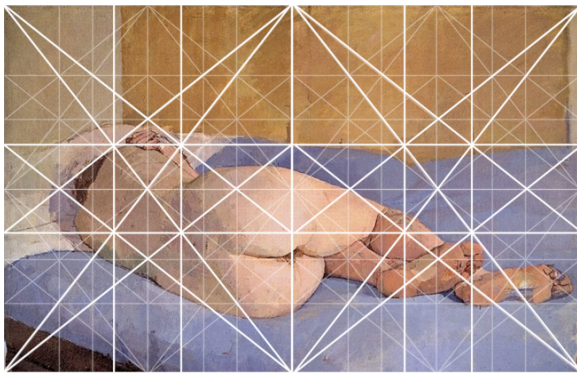
Now we can see Bouguereau using stacked 1.5 rectangles.



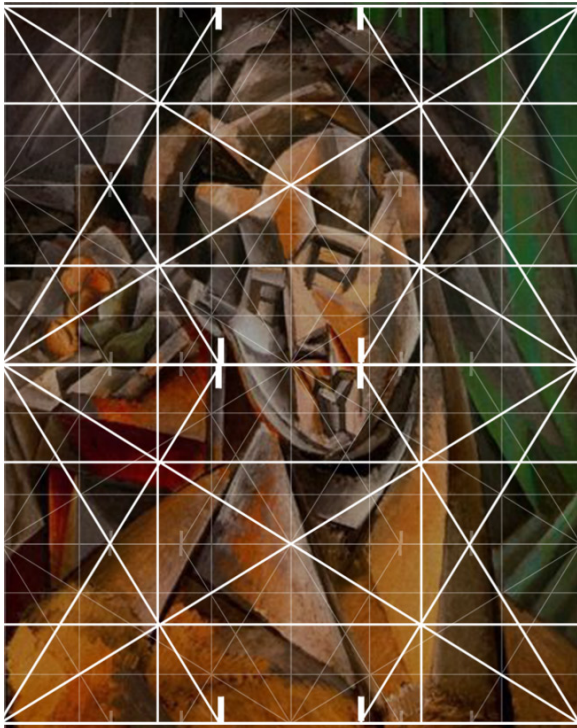
This is Gregory Crewdson using two root phi rectangles.



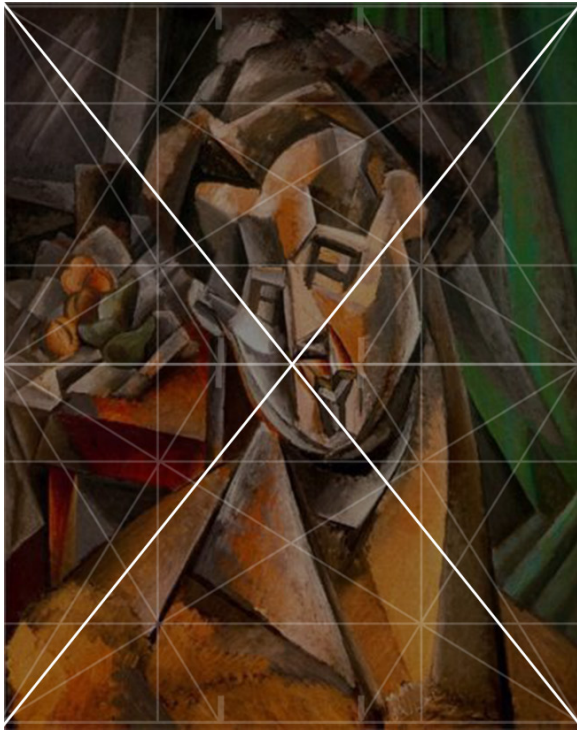
Euan Uglow using two root phi rectangles.



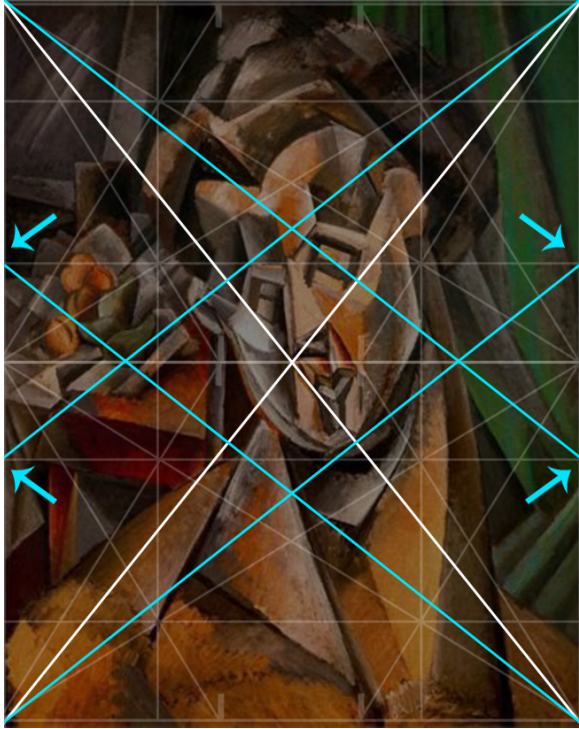
Picasso uses two Phi rectangles stacked.



If you are wondering, "Hey, what about the major diagonals of the whole canvas?" Nope, they aren't apparent when making compound root rectangles, but this doesn't mean you can't create them. Just run one from corner to corner if you wish to use that particular angle, it's as easy as that!



When we make the reciprocals from these two diagonals we see something very cool happen. They meet up with the lines of the stacked root 3 rectangles! As I mentioned, it's all about math, and unfortunately I can only explain it visually. Just as we see in Jay Hambidge's book, these rectangles have mathematical relationships which may be hard to explain, but give artists many more options.



As you can see from the examples provided, many masters were using this technique of compound root rectangles to help bring order to a canvas size that was not a standard root rectangle. I will continue to update this collection as I find more. Until then, I hope this helps some of you understand the versatility that can be found within root rectangles.